S.S.V.P.S's B. S. Deore Polytechnic, Dhule

**Department of Computer Engineering**

*Academic Year : 2025 – 2026*

# Presentation on Capstone Project Execution (CPE–316004)

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| **Topic** | Roommate Finder | |
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# Abstract


- The Roommate Finder System is a web-based platform developed to help people find rooms, flats, and compatible flatmates easily. In many cities, students and working professionals face difficulties while searching for rental rooms or trustworthy roommates. Most people rely on social media or brokers, which can be time-consuming and sometimes unreliable.
- The main goal of this system is to simplify the process of finding accommodation and flatmates while reducing the dependency on brokers. It also improves communication between people who are looking for shared accommodation and those who have rooms available.

# Content

1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

# 1. Introduction

The Roommate Finder System is a web-based platform designed to help people find rooms, flats, and suitable flatmates easily and quickly. In many cities, students and working professionals face difficulties while searching for rental rooms or compatible roommates. Usually, people depend on brokers, social media groups, or personal contacts, which can be time-consuming and sometimes unreliable. Because of this, many people struggle to find safe and affordable accommodation.

This system provides a simple online solution

# 2. Literature Review

- Several digital platforms have been developed to help people find rooms and flatmates online. Applications and websites allow users to search for rental rooms and shared accommodation. These systems reduce the need for brokers and manual searching.
- They also provide room details, images, and contact information so that users can easily communicate with room owners or potential flatmates.
- However, many systems still lack efficient communication and proper management features, which can make the process of finding suitable accommodation difficult.

# 3. Scope of the project

- The Flatmate Finder System provides the following features:

  User registration and login

- Room or flat posting with description and images

  Room category selection (Single Room, Shared Room, Flat, PG)

  Room and flatmate search system

  Admin dashboard for managing users and posts

  Communication or contact system between users

- This system can be used by students, working professionals, and people who are searching for rooms, flats, or suitable flatmates in different cities.

# 4. Methodology

- **The system works through the following steps:**

1. User registers and logs into the system
2. User posts room or flat details with description and images
3. Room or flat details are stored in the database
4. Admin verifies the post
5. Post is displayed for users searching for rooms or Roommate
6. Interested users contact the room owner or

# 4. Methodology

Fig. 4.1 flowchart

# 5. Details of designs, working & processes

- **Frontend Technology**
  - React JS
  - HTML
  - CSS
  - JavaScript
- **Backend Technology**
  - Node.js
  - Express.js
- **Database**
  - MongoDB

# 5. Details of designs, working & processes

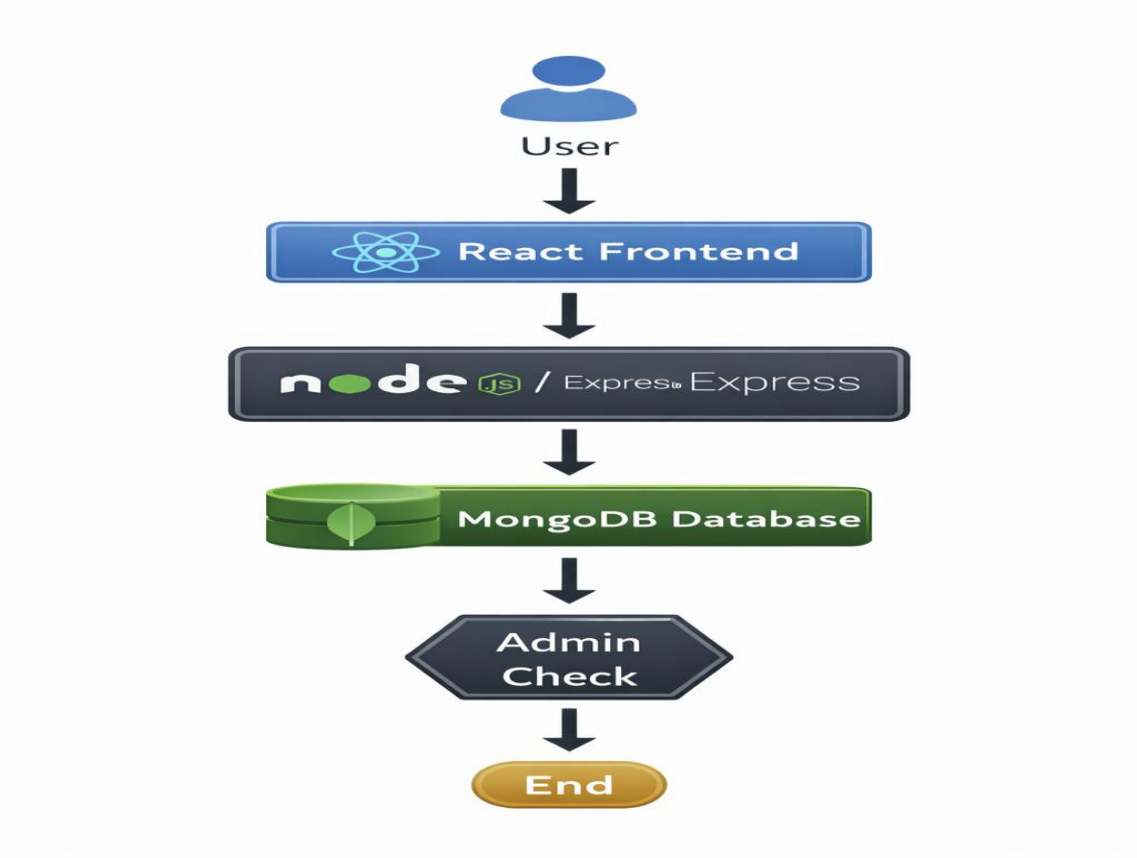


Fig. 5.1 System Architecture Diagram

# 5. Details of designs, working & processes

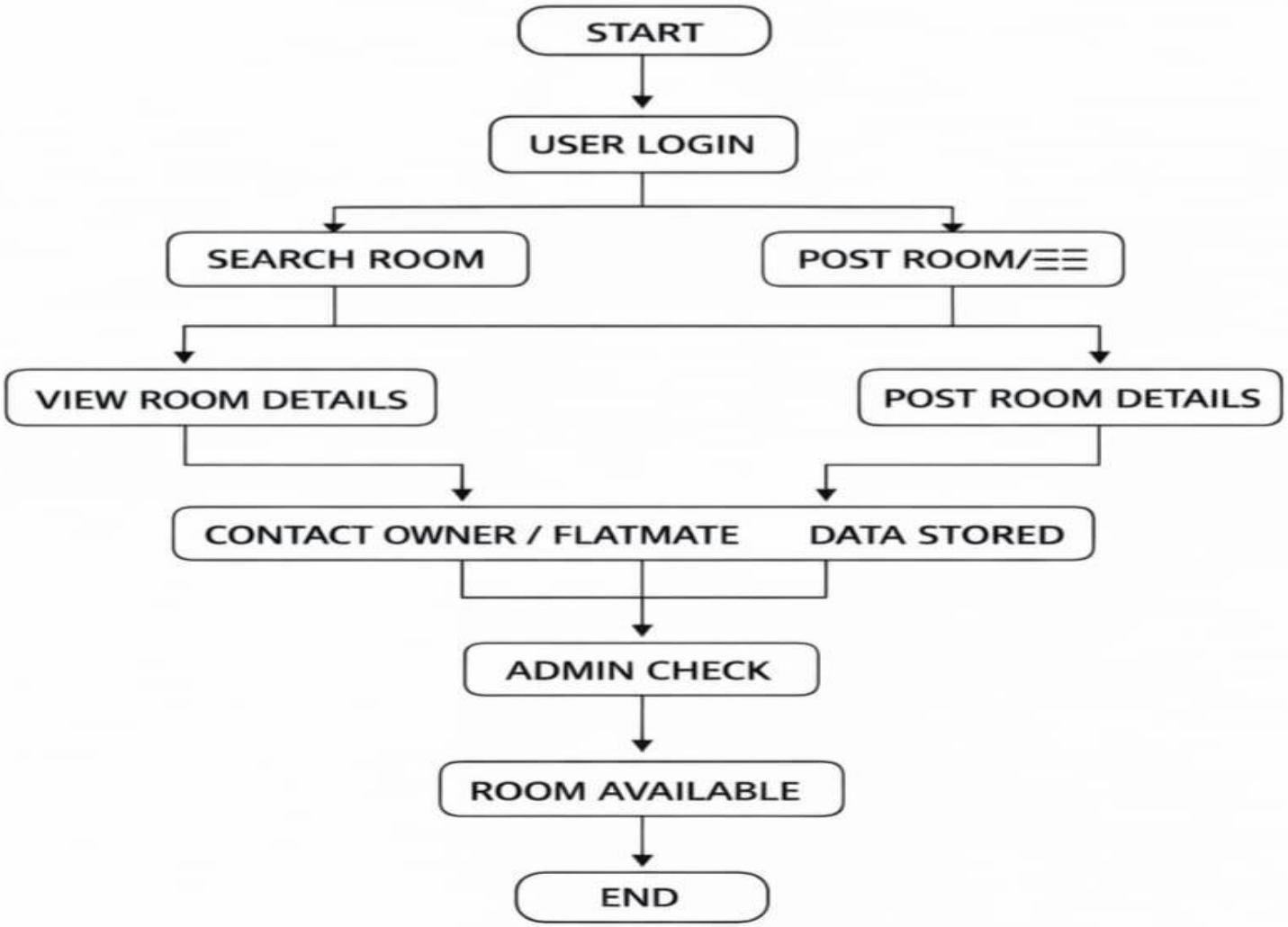


Fig 5.2 Flow of operations

# 6. Results and Applications

- **Results:**
- Faster room and flatmate searching
- Improved transparency of room details
- Reduced dependency on brokers
- **Applications:**
- Students searching for rooms
- Working professionals looking for roommates
- People who want to share rooms or flats

# 6. Results and Applications

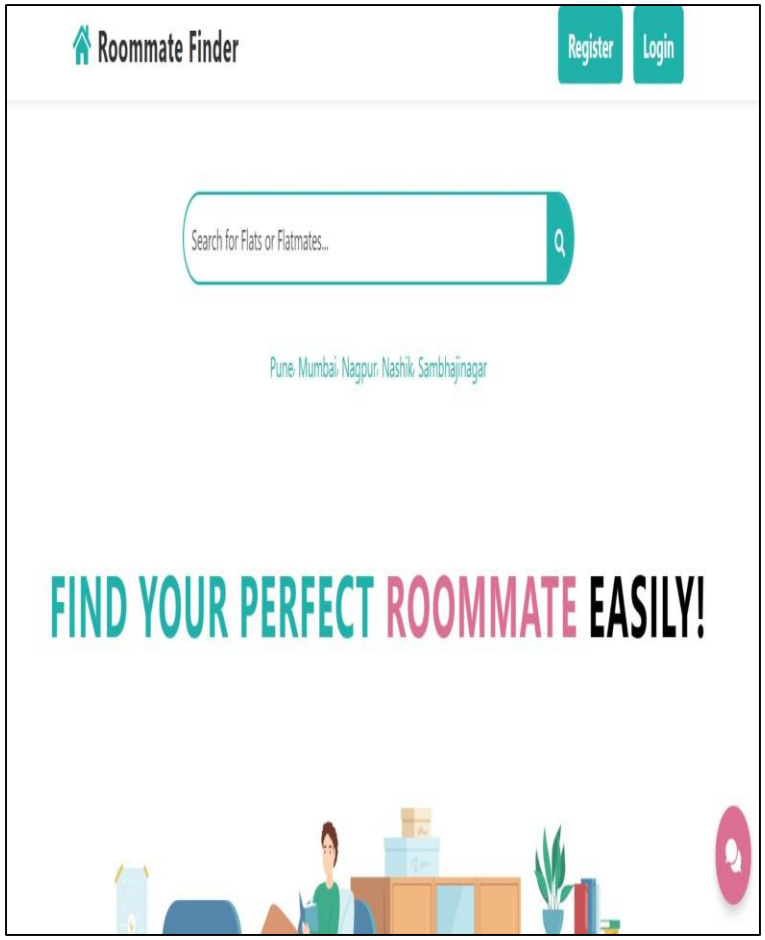


Fig. 6.1 Home page

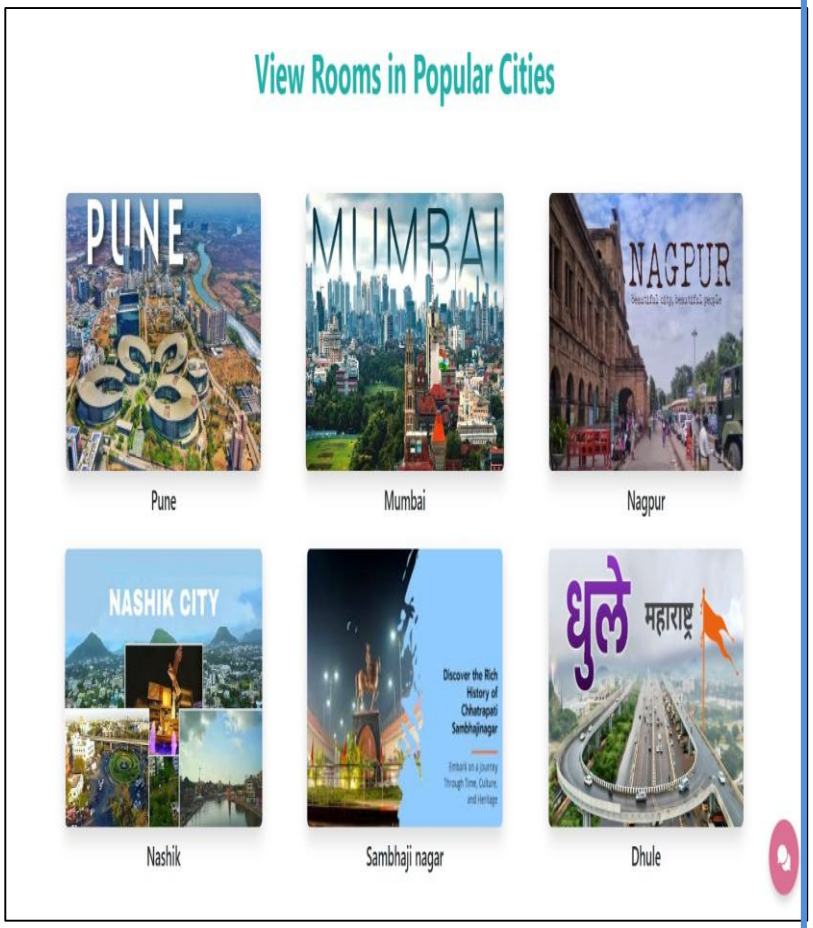


Fig 6.2 city listing page

# 6. Results and Applications

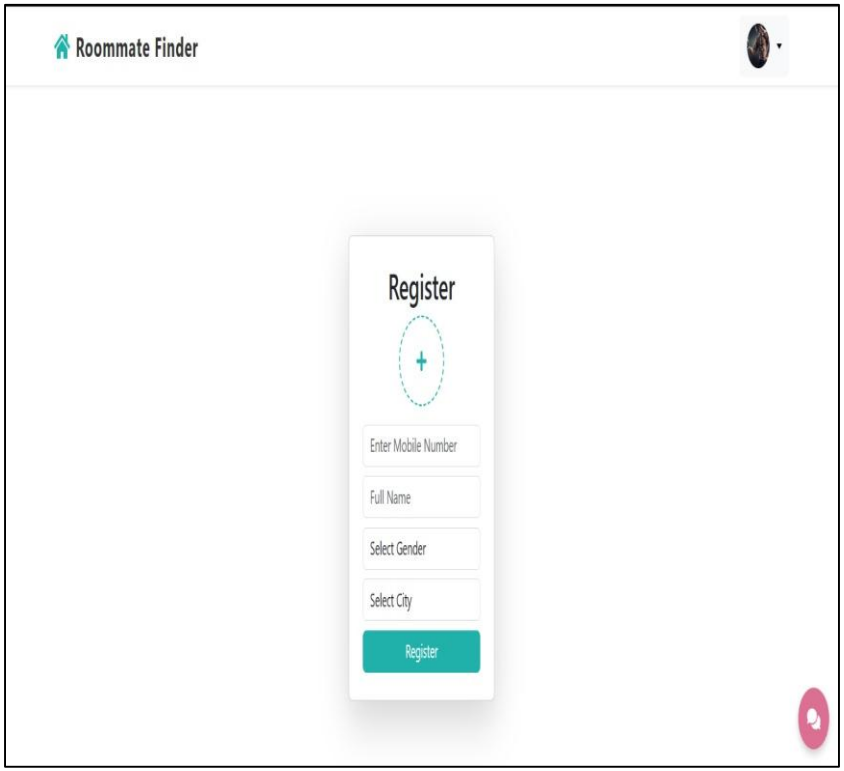


Fig. 6.3 User Registeration Interface

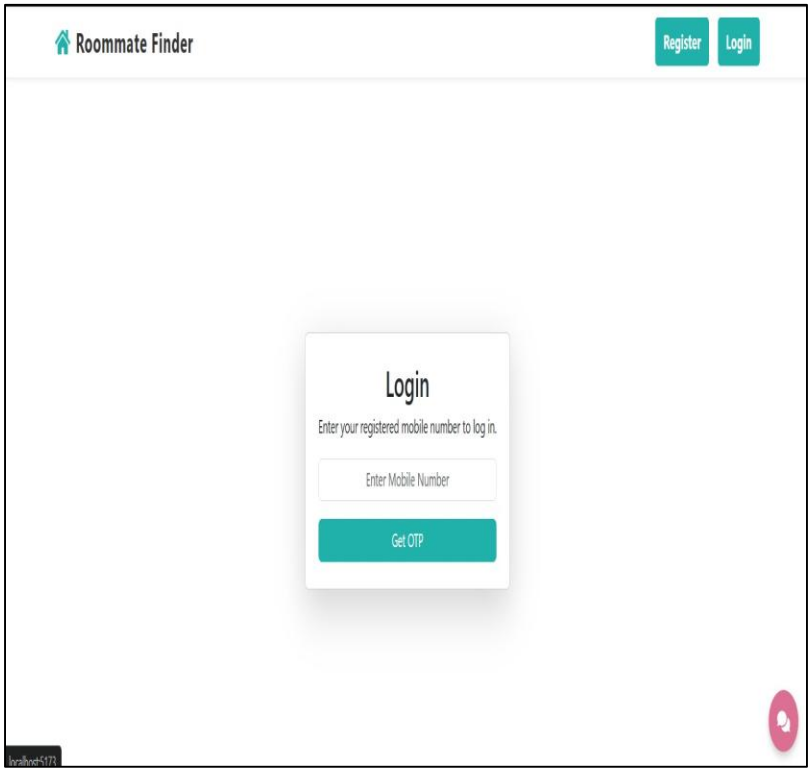


Fig 6.4 User Login page

# 6. Results and Applications

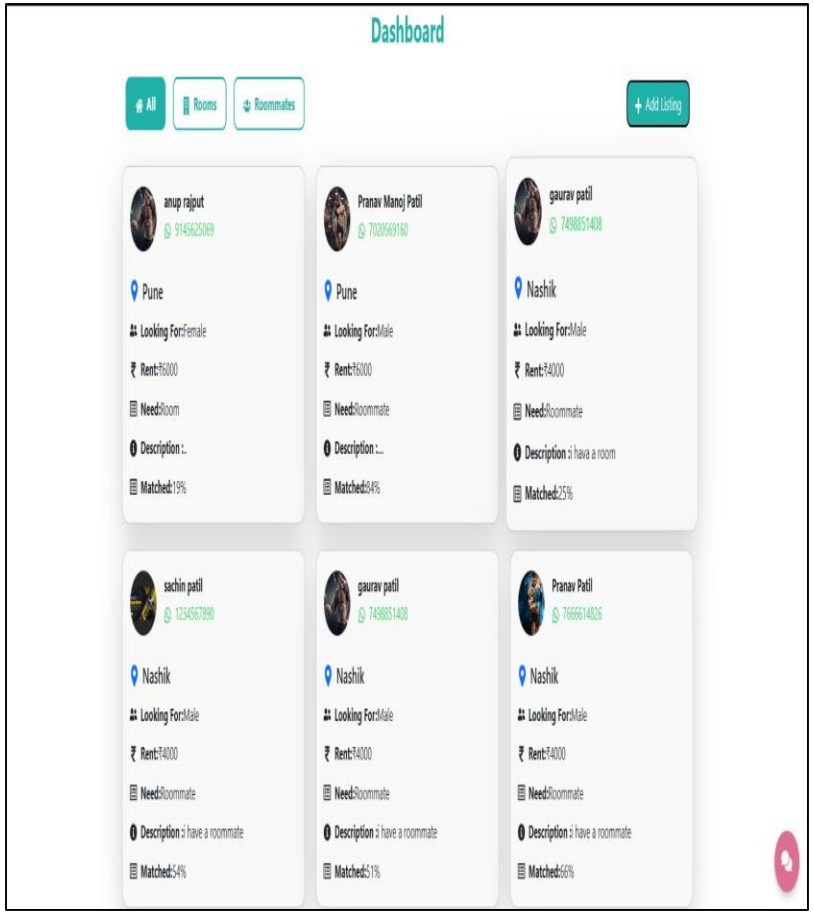


Fig 6.5 User Dashboard

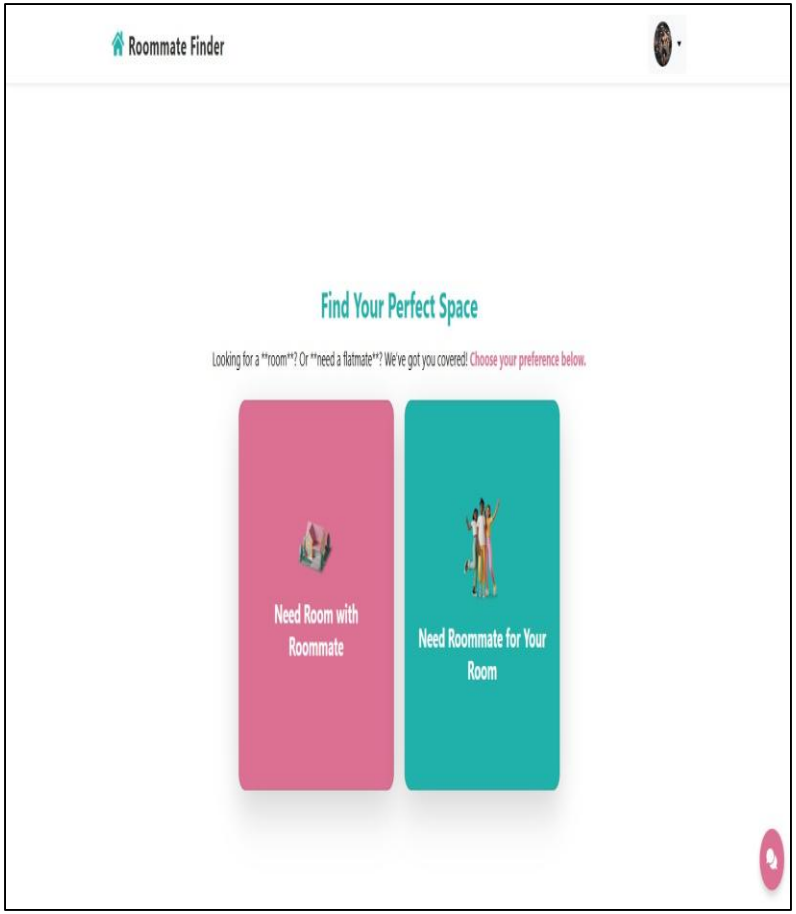


Fig 6.6 preference selection

# 6. Results and Applications

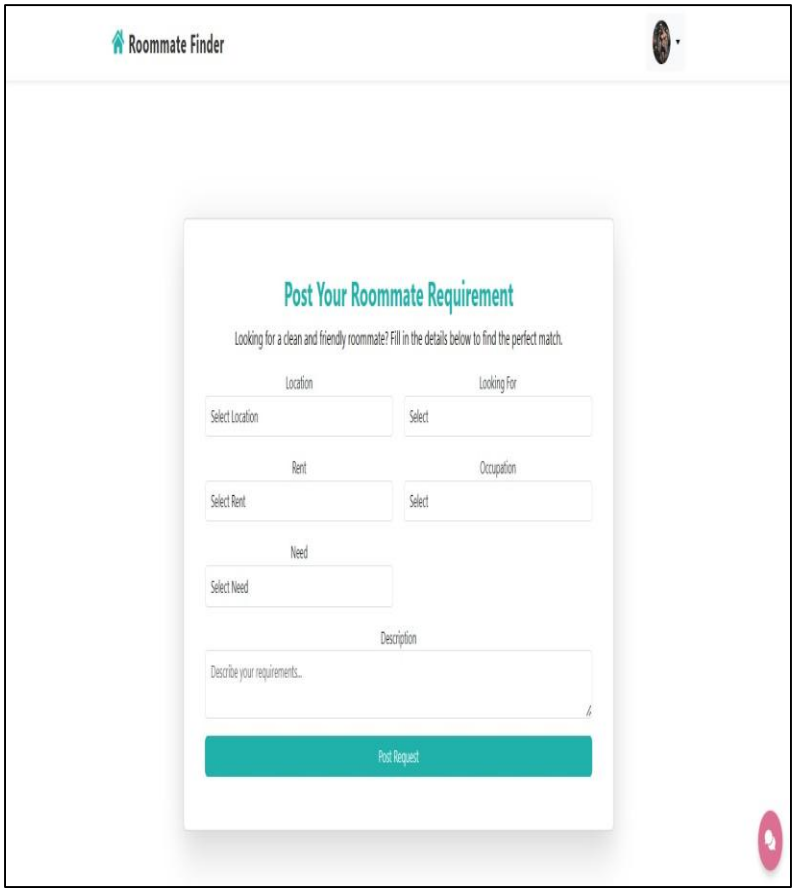


Fig. 6.7 Requirement form

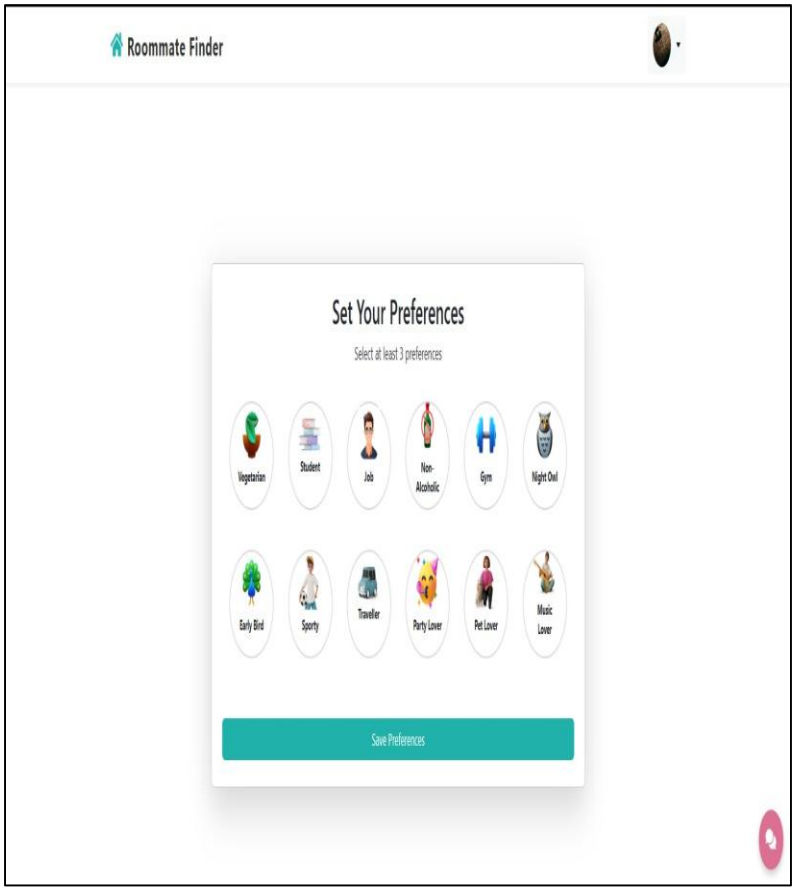


Fig. 6.8 Match & discovery Dashboard

# 6. Results and Applications

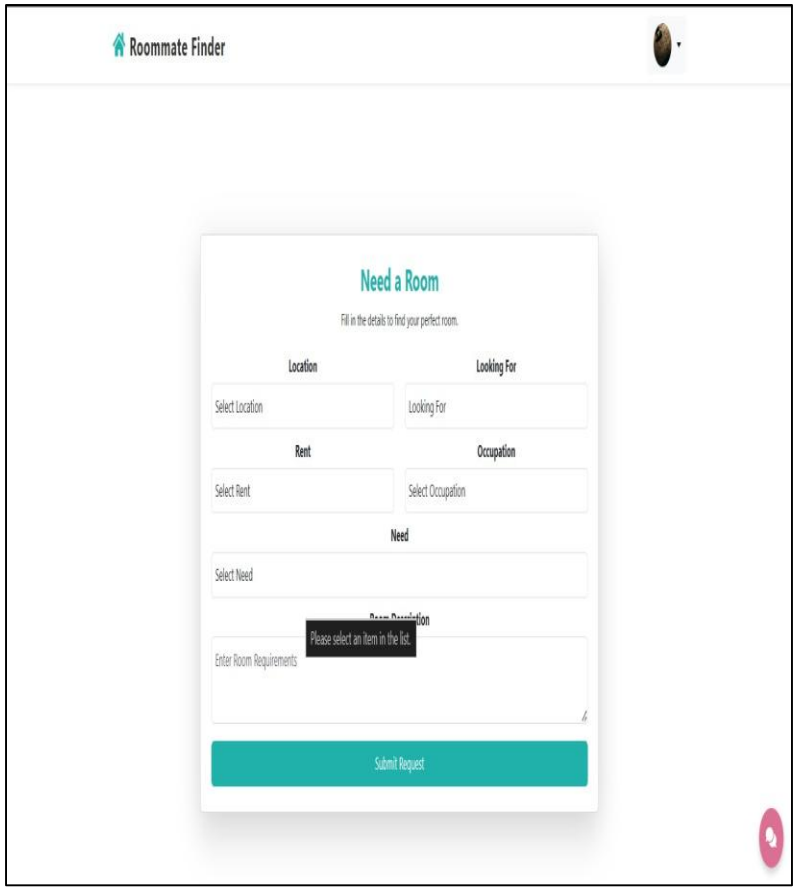


Fig. 6.9 Room Requirement

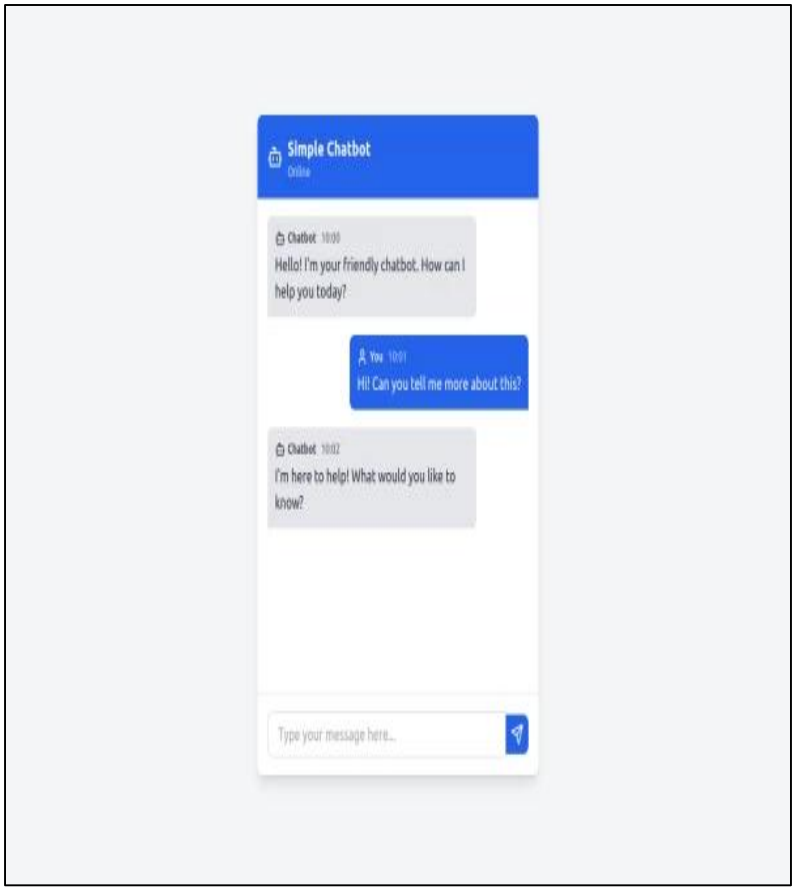


Fig. 6.10 User chatbox

# 7. Conclusion

- The Roommate Finder System helps people find rooms, flats, and suitable roommate in a simple and organized way.
- This system reduces the need for brokers and manual searching and provides a digital platform where room and roommate information can be easily shared and viewed.
- Through this system, users can search rooms, post room availability, and communicate with other users efficiently.
- Overall, the system simplifies the process of finding accommodation and helps people get suitable rooms or flatmates quickly and easily.

# 8.future scope

- Adding a dashboard for managing room and flat posts directly
- Implementing preference-based room and roommate search for better matching
- Adding a rating system where users can rate rooms or roommate after staying
- Integration with map services to visualize room or flat locations
- Using data analytics to identify frequently searched areas for rooms and shared accommodation

# 8. References and Bibliography

- **References**
- https://www.stanzaliving.com/
- https://www.w3schools.com
- https://www.geeksforgeeks.org/
- **Bibliography**
- Online resources and tutorials from web development websites and documentation.
- Research and references from the internet and educational materials related to roommate finder systems.